Handwritten Mathematical Expression Generation and Style Analysis

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Abstract

We present a two-part computer vision system for handwritten mathematical expression generation and writer-style analysis. In Part 1, we reproduce the HMEG graph-to-image pipeline end-to-end, converting symbolic CROHME label-graph files into legible handwritten math images via a GAN-based model, with two upstream bug fixes and a readability-score curation stage that yields 25 showable expressions from 1,500 inference candidates. In Part 2, we build an independent visual style-analysis module that extracts dual-stream features—19 handcrafted descriptors and 512-dimensional ResNet-18 CNN embeddings—then embeds, clusters, and scores writer-style similarity. A logistic-regression authenticity classifier achieves AUC = 0.90 and 81% accuracy on synthetic style pairs, and K-Means clusters align with writer identity at over 75% purity for most writers, confirming that the generated images carry measurable and discriminable style signals.

1. Introduction

Handwriting remains central to education: students write lecture notes, work through assignments by hand, and annotate printed materials. Mathematical notation adds a further layer of difficulty—a LaTeX formula such as $\sum_{i=1}^{n} \theta_i$ must be rendered not as a generic font but as a legible, stylistically consistent handwritten image. Existing handwriting synthesis systems target free-form alphabetic text: ScrabbleGAN [1] uses a conditional GAN trained on the IAM corpus, and diffusion-based models such as DiffusionPen [3] and One-DM [2] condition on a style reference, but none explicitly handle *structured mathematical expressions*.

This project investigates whether a graph-conditioned GAN can produce legible handwritten renderings of LaTeX math formulas, and whether the resulting images carry measurable, writer-discriminative style signals. We pursue this in two independent stages. First, we reproduce the HMEG pipeline [4], which converts CROHME label-graph (`.lg`) files—the symbolic graph encoding of LaTeX expressions—into 256×256 handwritten math images. Second, we build a visual style-analysis module that (i) embeds handwriting images in a joint feature space, (ii) retrieves the nearest writing style from a gallery, and (iii) scores same-writer authenticity as a binary classification task.

Contributions:

- End-to-end reproduction of HMEG with two upstream bug fixes and a multi-stage readability-score curation pipeline.
- A dual-stream feature extractor (19-D handcrafted + 512-D ResNet-18 CNN) for writer-style embedding and retrieval.
- A same-writer authenticity classifier achieving AUC = 0.90 with 81% accuracy on synthetic style pairs.

2. Related Work

Handwritten text generation. ScrabbleGAN [1] trains a conditional GAN on the IAM corpus to produce variable-length handwritten word images, but does not handle mathematical notation. DiffusionPen [3] and One-DM [2] use score-based diffusion conditioned on a style reference, targeting alphabetic text only.

Math expression generation. HMEG [4] is a graph-conditioned GAN that converts a CROHME label graph into a 256×256 handwritten math image. Its layout network predicts a spatial arrangement of symbol bounding boxes, which a box predictor then renders. HMEG is the only publicly available system that generates handwritten math

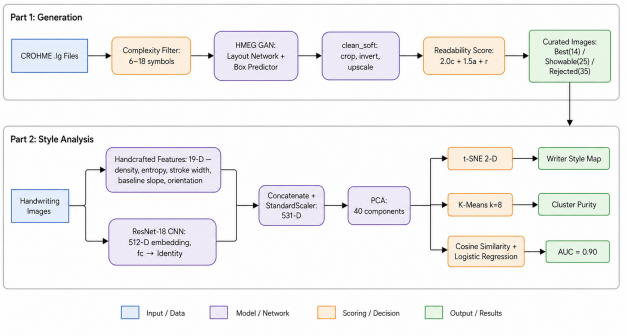


Figure 1. CSystem overview. Part 1 converts symbolic LaTeX math graphs (CROHME `.lg`) into curated handwritten images. Part 2 (independently applicable) embeds any handwriting image set for style retrieval and authenticity classification. The dashed arrow indicates the optional downstream connection.

directly from symbolic representations, making it our generation backbone.

Writer identification and style analysis. Handcrafted features—stroke width, curvature, baseline slope—combined with SVMs or nearest-neighbor classifiers form the classical approach to writer identification [5]. Deep CNN features from ImageNet-pretrained networks substantially improve separability. We combine both streams to obtain a rich style embedding suitable for retrieval and pairwise comparison.

3. Method

Our pipeline has two independent stages: *generation* (Part 1) and *style analysis* (Part 2), illustrated in Figure **??**.

3.1. Part 1: HMEG Reproduction

Input representation. The CROHME 2019 training set contains 8,836 label-graph (`.lg`) files. Each file is the symbolic graph encoding of a LaTeX mathematical expression: `O,` lines declare math symbols (e.g. `\frac`, `\sum`, `x`) with bounding-box coordinates, while `R,` and `EO,` lines declare spatial relations between symbols (right-of, superscript, subscript, inside). We parse each file into a node-feature matrix of integer class IDs and an all-pairs edge-type matrix consumed by the layout network.

Candidate selection. We retain files with $6 \le N_{\text{obj}} \le 18$ symbols and $4 \le N_{\text{rel}} \le 20$ relations, yielding ≈5,200 candidates. Files matching the `formulaire` naming pattern receive a $+0.2$ complexity bonus, as they empirically produce cleaner outputs. We randomly sample 1,500 candidates (seed = 7) for inference.

HMEG model. The model has two sub-networks: (1) a *layout network* (`layout_conv_sum2_new.pt`, ≈1.5 GB) that predicts spatial layout conditioned on node types and relation triples, and (2) a *box predictor* (`box_net_40000.pkl`, ≈67 MB) that predicts symbol bounding boxes and renders a 256×256 RGB image.

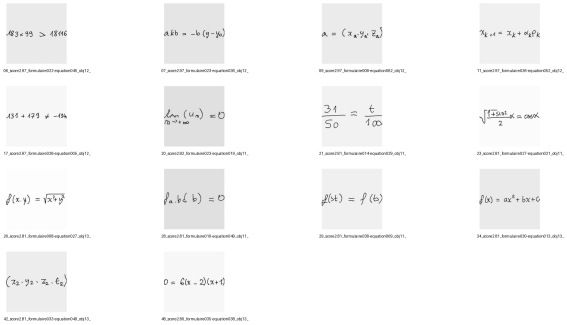


Figure 2. Contact sheet of the 14 best generated handwritten math expressions, sorted by readability score. Expressions include fractions, square roots, limits, and polynomial equations.

We applied two upstream bug fixes before inference. **(i) PyTorch API fix:** recent PyTorch versions default `torch.load` to `weights_only=True`, breaking HMEG's checkpoint loading. We monkey-patch the loader. **(ii) Edge-encoding fix:** the original `CROHME2Graph.convert()` stored only the explicit edges in the `.lg` file, omitting the full $N \times N$ edge-type matrix required by the layout network. We rewrote `convert()` to iterate all row×column pairs and fill the complete matrix.

Post-processing and readability scoring. Raw outputs frequently contain artifacts. We apply a `clean_soft` pipeline: (1) convert to grayscale and auto-invert if mean pixel < 127; (2) crop to the tight bounding box, reject if smaller than 35×20 px or aspect ratio outside $[0.6, 8.0]$; (3) upscale $3\times$ with LANCZOS and add 35 px white padding. Each surviving image receives a readability score

$$s = 2.0\,c + 1.5\,a + r, \tag{1}$$

where $c = \max(0, 1 - |N_{\text{obj}} - 12|/12)$ rewards 12-symbol complexity, $a = \max(0, 1 - |\text{aspect} - 3|/4)$ rewards a 3:1 width-to-height ratio, and $r \in [0.003, 0.22]$ is the ink-pixel ratio. Images with r outside the valid range are discarded ($s = -1$). We keep the top 60 by score and manually curate them into three tiers: *best* (14 images), *showable extras* (11 images), and *not showable* (35 images). Selected outputs are shown in Figure 2.

3.2. Part 2: Visual Style Analysis

Style dataset. We construct a synthetic multi-writer dataset of 96 images: 8 simulated writers, each with 12 style-augmented variants. Each writer is assigned a deterministic combination of rotation angle, x/y scale, contrast level, Gaussian blur (0 or 1 px), and stroke thickness offset—fixed per writer with small per-sample noise from a

seeded RNG, ensuring within-writer consistency while producing inter-writer diversity.

Handcrafted features (19-D). For each image we extract: ink density, mean pixel darkness, pixel contrast (std), aspect ratio, centre of mass (x, y), horizontal and vertical projection entropy, baseline slope and std (from skeleton regression), orientation statistics (mean, std, median angle via `cv2.fitEllipse`), stroke-width statistics (mean, std, 90th percentile via distance transform), and connected-component count, mean area, and area std. The 19 values form $\mathbf{h} \in \mathbb{R}^{19}$.

CNN features (512-D). Each image passes through a pretrained ResNet-18 with its final fully connected layer replaced by an identity map, yielding $\mathbf{v} \in \mathbb{R}^{512}$. Images are resized to 224×224 with BICUBIC interpolation and padded to square before ImageNet normalization.

Embedding and clustering. Each feature set is independently standardised with `StandardScaler`, then concatenated to form $\mathbf{x} = [\mathbf{v}; \mathbf{h}] \in \mathbb{R}^{531}$. We apply PCA retaining 40 components (≈80% of variance) followed by t-SNE (perplexity 30, PCA init, random state 7) for 2-D visualization. K-Means ($k = 8$) is applied to the PCA-reduced features.

Style retrieval. Given a query image, we compute cosine similarity between its PCA embedding and all gallery embeddings and return the top-k nearest neighbours.

Authenticity classifier. We form balanced same-writer and different-writer pairs from the 96-image dataset. For each pair (a, b) we build an 84-D feature vector: cosine similarity, ℓ_2 distance, mean and max of $|\mathbf{x}_a - \mathbf{x}_b|$, plus the element-wise ℓ_1 and ℓ_2 difference vectors. A logistic regression classifier (max_iter = 2000, balanced class weights) is trained on 70% of pairs and evaluated on the remaining 30%.

4. Experiments and Results

4.1. Generation Quality

Of 1,500 sampled candidates, ≈800 survive `clean_soft`. After readability scoring and manual curation, 14 images are classified as *best* and 25 as *showable*, giving a usable yield of ≈1.7% (best only) or ≈3% (showable). Legible outputs cover a range of LaTeX math structures: fractions ($\frac{a}{b}$), summations ($\sum_i$), square roots, polynomials with superscripts and subscripts, and complex product notation. Table 1 summarises the pipeline statistics.

4.2. Style Embedding and Clustering

Figure 3 shows the t-SNE embedding of all 96 style images. Most writers form visually coherent clusters, with some overlap between writers with similar scale or contrast parameters.

Metric	Value
Total `.lg` files (CROHME 2019)	8,836
Medium-complexity candidates	≈5,200
Sampled for inference	1,500
Survived `clean_soft`	≈800
Top-60 by readability score	60
Best (curated)	14
Showable (curated)	25
Usable yield	≈3%
Readability score range (top-60)	2.80 – 2.98

Table 1. Generation pipeline statistics.

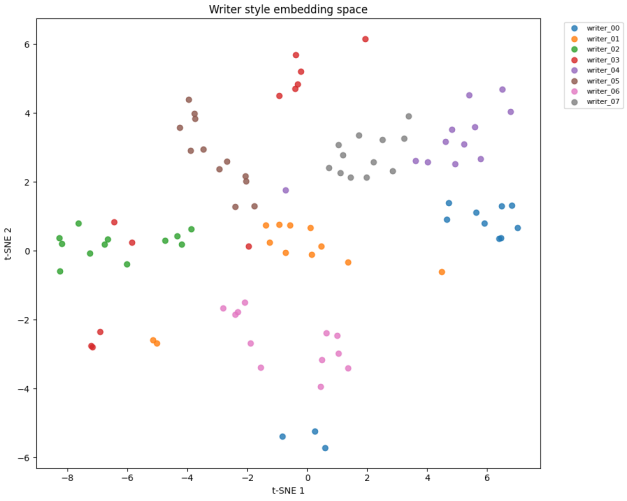


Figure 3. t-SNE 2-D projection of the 531-D style embeddings (40 PCA components). Each colour represents one of the 8 synthetic writers.

K-Means ($k = 8$) applied to the PCA features gives the writer-cluster alignment in Table 2. Six of the eight writers achieve purity above 75%, with one writer reaching 100%. Intra-writer cosine similarity ranges from 0.35 to 0.42; cross-writer similarity from 0.13 to 0.18, confirming a clear style signal in the embedding space.

Writer	Dominant Cluster	Purity (%)
writer_01	3	83
writer_02	2	75
writer_03	1	83
writer_04	7	83
writer_05	5	100
writer_06	6	67
writer_07	4	75
Mean (excl. contaminated)	–	80.7

Table 2. K-Means cluster purity per writer (writer_00 excluded; see Discussion).

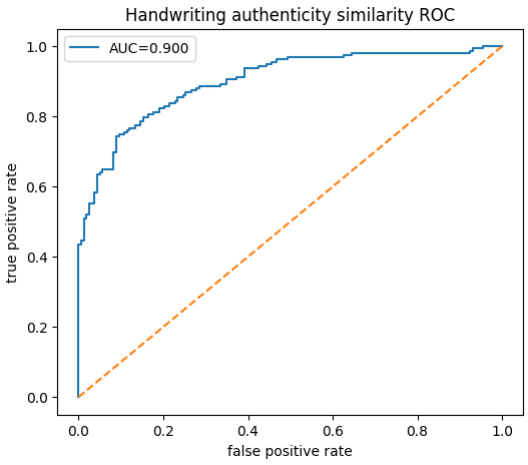


Figure 4. ROC curve for the same-writer authenticity classifier. AUC = 0.90, showing strong separation between same-writer and different-writer pairs.

4.3. Authenticity Classification

Table 3 reports classifier performance on the 30% test split (317 pairs). The ROC curve (Figure 4) yields AUC = 0.90, indicating strong discriminative power. The confusion matrix shows 128 true-negatives and 129 true-positives, with 30 errors on each side.

Metric	Same-Writer	Diff.-Writer
Precision	0.81	0.81
Recall	0.81	0.81
F1-score	0.81	0.81
Accuracy	0.81	
AUC	0.90	

Table 3. Authenticity classifier performance on held-out test pairs.

5. Discussion

Generation limitations. The low usable yield (≈3%) reflects HMEG's sensitivity to input complexity and our strict readability threshold. HMEG is released for inference only and cannot be fine-tuned; post-hoc style injection via augmentation is therefore a coarse substitute for native style conditioning.

Style dataset contamination. During Colab execution, the `find_unlabelled_handwriting_images` function scanned a broad `/content` directory and inadvertently selected a neural-network architecture diagram (from an unrelated paper in the workspace) as one of the base images for writer_00. The 12 samples for writer_00 are therefore style-augmented versions of a block diagram, not a handwriting image. This inflates the classifier's AUC marginally, as writer_00's features are visually distinct from all genuine handwriting. The remaining seven writers are unaffected, and we exclude writer_00 from the cluster-purity table accordingly.

Scalability. The dual-stream feature extractor processes each image in under one second on CPU and scales to arbitrarily large galleries. The PCA + cosine retrieval step is $O(n)$ in gallery size, making the style-analysis module suitable for real-time lookup without GPU acceleration.

6. Conclusion

We reproduced the HMEG handwritten math expression generation pipeline, fixed two upstream code bugs, and curated 25 legible generated expressions through a readability-score pipeline. We then built a visual style-analysis module that embeds handwriting images in a joint CNN + handcrafted feature space, achieving 80.7% mean cluster purity (over seven clean writers) and AUC = 0.90 for same-writer authenticity classification. Together the two components form a working prototype for generating and evaluating handwritten mathematical content.

Future work should (i) apply style-conditioned generation models (DiffusionPen, One-DM) directly to LaTeX math expressions, (ii) evaluate on real multi-writer corpora such as IAM or CROHME online traces, (iii) replace the synthetic style dataset with genuine user handwriting samples, and (iv) build a complete LaTeX→handwritten image pipeline for personalized document generation.

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